

Received 12 May 2025, accepted 19 May 2025, date of publication 26 May 2025, date of current version 3 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3573443

RESEARCH ARTICLE

Eliminating Meteorological Dependencies in Solar Power Forecasting: A Deep Learning Solution With NeuralProphet and Real-World Data

NECATI AKSOY^{1,2}, ALPER YILMAZ^{1,2}, GÖKAY BAYRAK^{1,2}, AND MEHMET KOÇ^{3,4}

¹Renewable Energy Systems and Smart Grids Laboratory, Department of Electrical and Electronics Engineering, Bursa Technical University, 16300 Bursa, Türkiye

²Electric Vehicles Application and Research Center, Bursa Technical University, 16300 Bursa, Türkiye

³Uludağ Elektrik Dağıtım A.Ş. (UEDAŞ), 16310 Bursa, Türkiye

⁴Graduate School, Bursa Technical University, 16310 Bursa, Türkiye

Corresponding author: Necati Aksoy (necati.aksoy@btu.edu.tr)

ABSTRACT Forecasting solar power generation is essential for efficient energy management and grid stability. However, existing predictive models often rely on external datasets, such as meteorological and sensor data, to make accurate predictions. This dependency introduces complexities and limits their application in data-sparse scenarios. In this study, we propose a novel forecasting approach based on the NeuralProphet algorithm, a deep learning model that predicts solar power generation solely from its historical data, eliminating reliance on additional input data. To evaluate the proposed approach, we conducted two case studies. The first utilized a 10-month dataset from a 1.2 kW small-scale solar power unit at Bursa Technical University's Smart Grids laboratory, recorded at 15-minute intervals. Despite the limited dataset, the model achieved an R-squared value exceeding 0.74, demonstrating promising predictive capability. The second case study applied the NeuralProphet-based model to a large-scale dataset of nationwide solar power generation in Germany, spanning five years and collected at 15-minute intervals. Models trained on this dataset achieved R-squared values exceeding 0.99, highlighting the algorithm's capacity to effectively capture seasonal and temporal patterns at a national scale. Our results indicate that the NeuralProphet-based forecasting approach offers a viable and efficient alternative for solar power prediction, achieving high accuracy without external data dependencies.

INDEX TERMS Deep learning, NeuralProphet, predictive models, renewable energy, solar power forecasting.

I. INTRODUCTION

The accurate forecasting of solar power generation is critical for efficient energy management, grid stability, and the integration of renewable energy sources into existing power systems. Conventional forecasting methods often rely on diverse input data, including meteorological measurements, grid data, and other sensor-based information, to achieve high predictive accuracy. While effective, these methods present significant challenges, including increased complexity, dependency on external data sources, and reduced

applicability in regions or scenarios where such data is unavailable or unreliable. These limitations underline the need for alternative forecasting approaches that can operate independently of supplementary data while maintaining robust performance.

In response to these challenges, this study proposes a novel deep learning-based forecasting approach utilizing the NeuralProphet algorithm. NeuralProphet is a time-series forecasting model that leverages seasonality, trend, and temporal dependencies within historical data. Its ability to extract patterns solely from the target variable makes it uniquely suited for forecasting solar power generation, which exhibits strong seasonal and temporal regularities.

The associate editor coordinating the review of this manuscript and approving it for publication was Yun Seng Lim¹.

TABLE 1. Abbreviations and symbols used throughout the paper.

Abbreviation	Definition	Symbol	Description
AI	Artificial Intelligence	y_t	Actual solar power generation at time t
AR	Auto-Regressive	$\hat{y}_t$	Forecasted solar power generation at time t
BTU	Bursa Technical University	$T(t)$	Trend component of the time series
CNN	Convolutional Neural Network	$S(t)$	Seasonal component
DL	Deep Learning	$E(t)$	Event or holiday effect
GBM	Gradient Boosting Machine	$F(t)$	Future regressor contribution
GRU	Gated Recurrent Unit	$A(t)$	Auto-regressive effect
LSTM	Long Short-Term Memory	$L(t)$	Lagged regressor component
MAE	Mean Absolute Error	δ_0	Initial growth rate of trend
MAPE	Mean Absolute Percentage Error	ρ_0	Initial offset of trend
MBE	Mean Bias Error	$\Gamma(t)$	Binary indicator vector for changepoints
ML	Machine Learning	p	Periodicity (e.g., daily, weekly, yearly)
NMAE	Normalized Mean Absolute Error	k	Number of Fourier terms
PV	Photovoltaic	a_j, b_j	Coefficients of Fourier seasonal terms
RMSE	Root Mean Squared Error	$x \in X$	Set of lagged regressors
RNN	Recurrent Neural Network	$f \in F$	Set of future regressors
SMAPE	Symmetric Mean Absolute Percentage Error	$e \in E$	Set of events/holidays
TSO	Transmission System Operator	n	Number of observations in the dataset
		β	Threshold in Huber loss function
		θ	Model parameter vector
		ϵ, α	Regularization constants
		$\bar{y}$	Mean of actual values
		SS_{RES}	Sum of squares of residuals
		SS_{TOT}	Total sum of squares

Unlike traditional models, the proposed method eliminates the reliance on external datasets, simplifying the forecasting process and broadening its applicability to data-sparse environments.

To validate the proposed approach, two distinct case studies were conducted. The first involved a small-scale solar power generation unit in the Smart Grids laboratory (lab.) of Bursa Technical University (BTU), where 10 months of solar power generation data were collected at 15-minute intervals. The second case study examined large-scale solar power generation across Germany, utilizing five years of nationwide data collected by the country's Transmission System Operators (TSOs). These datasets allowed for a comprehensive evaluation of the model's performance across varying scales and data availability conditions. To provide a comprehensive understanding of the proposed approach, the overall workflow of the study is illustrated in Figure 1.

This paper is organized as follows: Section II reviews related works in solar power forecasting, focusing on existing methods and their limitations. Section III describes the data acquisition process and dataset preparation steps undertaken for both case studies. Section IV explains the NeuralProphet algorithm, detailing its mathematical foundations and the proposed forecasting methodology. Section V presents the performance results and analysis, comparing the proposed model's outcomes against established metrics. Finally, Section VI concludes with a summary of the study's contributions, limitations, and directions for future research. Furthermore, the meanings of the abbreviations and symbols used throughout this paper are also given in Table 1.

II. RELATED WORKS

In the current literature, solar energy forecasting solutions typically rely on a range of external meteorological variables,

such as irradiance (which may include direct, diffuse, or ultraviolet components), ambient temperature, wind speed, cloud cover, and other environmental factors. While these approaches provide valuable insights, they often suffer from data dependency issues, where obtaining accurate and comprehensive meteorological data in real-time may not always be feasible or practical. Addressing these limitations necessitates the exploration of methods capable of producing accurate predictions using minimal and more accessible input data.

Artificial intelligence (AI) techniques have significantly contributed to system control, decision-making processes, future strategies, and efficiency improvements. To this end, numerous machine learning (ML) and deep learning (DL) algorithms have been integrated into renewable energy applications. Similar to the approach presented in this study, various ML algorithms have been employed for solar power forecasting using different training datasets. For instance, a two-stage forecasting study in [1] first predicted irradiance data from numerical inputs, and subsequently utilized the estimated irradiance to forecast solar power generation. Additionally, some studies combine multiple algorithms [2], while others use modified or enhanced versions of the same algorithm in conjunction with diverse scientific techniques [3]. Beyond ML methods, DL approaches have also been introduced for solar energy forecasting in several studies [4], [5].

In certain studies, deep or basic artificial neural networks (ANNs) were trained for predictions [6], whereas Long Short-Term Memory (LSTM) networks or their derivatives were sometimes preferred [7], [8]. Another critical factor influencing the choice of methodology is whether the forecasting task targets short-term or long-term predictions. Likewise, day-ahead forecasting studies have achieved successful

results [9]. Moreover, selecting appropriate training data significantly impacts the performance outcomes of these models. Some studies incorporate geographical features into the training process, with satellite imagery emerging as a popular choice for training data in this context [10]. Convolutional Neural Networks (CNNs), a powerful deep learning method, have demonstrated strong performance in this domain. Both LSTM and CNN are frequently combined with deterministic, heuristic, or metaheuristic optimization methods to improve learning performance [11]. However, a common challenge with these approaches lies in their high computational power requirements and extended training times.

Meteorological data, particularly irradiance, plays a pivotal role in influencing solar energy production. Consequently, certain studies have focused on irradiance forecasting rather than power forecasting and have achieved notably successful results [12], [13], [14]. To address the need for shorter training times, innovative regression algorithms based on Gradient Boosting Machines (GBM) have been investigated, yielding promising outcomes [15]. For instance, [15] proposed three novel GBM-based algorithms—XGBoost, CatBoost, and LightGBM—to predict the power output of a solar power plant, with comparative results showcasing their success. Similarly, in another study, the XGBoost algorithm demonstrated significant promise and robustness in ultra-short-term energy forecasting [16]. Furthermore, GBM-based methods have been integrated with DL techniques for wind turbine forecasting, as demonstrated in [17], where LightGBM combined with CNN proved to be an effective solution for wind turbine power generation prediction. In another similar study addressing the wind turbine power forecasting problem, two popular DL techniques, CNN and LSTM, were applied alongside LightGBM, resulting in a successful forecasting model [18].

Studies in the context of DL are in constant development. In this regard, [19] proposes a method combining Neural-Prophet and CNN-LSTM models to predict the next day's electricity load based on the past two days' load data. Data from Hong Kong (August 1, 2022 – June 15, 2023, 30-minute sampling, 15,312 data points) and Texas (January 1, 2023 – November 30, 2023, hourly sampling, 8,016 data points) were used. The model achieved high accuracy with MAE = 2.13/21.30, MAPE = 4.19%/2.04%, and RMSE = 2.57/30.05. The literature review discusses the performance of Prophet, LSTM, and hybrid models in load forecasting, highlighting that the proposed method outperforms these approaches. On the other hand, [20] proposes a convolutional LSTM-based framework for predicting solar irradiance from one day to one week ahead using historical irradiance data and weather parameters (temperature, humidity, and rainfall). The model uses irradiance data from the previous 3–5 days as input and incorporates weather forecast features for improved accuracy. The proposed model achieves a normalized mean absolute error (NMAE) of 25.3% for

day-ahead predictions and reduces errors by up to 36% compared to existing methods. A Dynamic Bayesian Network (DBN)-based framework is proposed for solar power generation forecasting in PV plants, utilizing meteorological data, operational indicators, sensor data (including IGBT temperatures, number of faulty inverter modules, inverter on/off status, and curtailment status), and historical power data. The model, applied to a dataset with 15-minute intervals from a 40 MW solar plant over a year, achieved 92%-95% accuracy for forecasting horizons ranging from one hour to one week. It surpassed benchmark methods such as SVR, kNN, ANN, and LSTM [21]. The study [22] proposes a short-term photovoltaic power output forecasting method using Random Forest (RF), Deep Neural Network (DNN), and LSTM models. The approach utilizes readily available meteorological data such as ambient temperature, humidity, wind speed, atmospheric pressure, and cloud coverage, along with historical PV output data from a 5.5 kW PV system in Berlin, Germany, collected over four years with hourly resolution. The RF model achieved the best performance, with a Mean Absolute Error (MAE) of 309.46 W and Root Mean Square Error (RMSE) of 443.42 W. On the other hand, deep learning algorithms hold promise for improving PV forecast accuracy and reliability but remain underutilized [23]. In [24], RNN, LSTM, GRU and Bi-LSTM forecasting models are extensively trained for solar power forecasting and remarkable results are presented. Among these models, the Bi-LSTM model outperforms the others, but the models used in the study by Aksoy et al. still require input data, unlike the novelty presented in this paper.

LSTM-based recursive and MIMO strategies are applied for PV power forecasting [25], leveraging a five-year hourly dataset from a 25 kW PV plant in Romania. The model predicts up to two days ahead, achieving RMSE of 1.91 kW (recursive) and 1.22 kW (MIMO), with MIMO demonstrating superior accuracy for longer horizons. The innovative aspects include analyzing the impact of data size, forecast horizon, and input configuration while comparing recursive and non-recursive strategies, addressing key gaps in multi-step forecasting. MLFFNN, RNN, and NARXNN to estimate and forecast PV output power, utilizing two months of data (surface temperature and solar radiation) from four real PV substations in Egypt [26]. The training process utilized 45 days of data, while the remaining 15 days were employed for validation, ensuring the generalization capability of the trained models under different conditions. The MLFFNN achieved the best performance with the lowest MSE (0.27533) and absolute approximation error (0.2099 MWh), demonstrating superior accuracy in both training and upscaling for regional power estimation. The review in [27] analyzes ML approaches for short-term solar PV power forecasting, comparing methods such as RF, Gradient Boost, and NNs using historical power generation data and meteorological inputs [27]. The Random Forest model achieved the best performance with a normalized

RMSE of 10.07% and a skill score of 37.33%, showcasing its ability to handle nonlinear relationships effectively.

Recently, novel deep learning architectures such as N-HiTS (Neural Hierarchical Interpolation for Time Series) and N-BEATS (Neural Basis Expansion Analysis for Time Series) have emerged as powerful alternatives for time-series forecasting. The N-HiTS model, in particular, is known for its flexible architecture and hierarchical design that improves long-range forecasting accuracy. In a recent study by [28], the N-HiTS model was successfully applied to short-term and long-term solar power prediction, showing superior accuracy when compared to classical models. Similarly, N-BEATS, a model that relies on a deep fully-connected architecture without requiring domain-specific input features, has also proven effective for forecasting solar energy production. According to [29], N-BEATS outperformed conventional LSTM and GRU models in solar generation prediction scenarios, particularly in cases involving limited input variables. These advancements suggest that N-HiTS and N-BEATS are promising additions to the deep learning toolkit for renewable energy forecasting and merit further exploration in practical applications.

One of the additional benefits of developing fast and highly accurate forecasting models, as targeted in this study, lies in their reinforcement of ML-based control applications. Parallel to the field of control, where system dynamics are learned from data and ML models, reinforcement learning (RL), which focuses on learning to control a system through an environment model, has been gaining increasing popularity. In the context of energy management or control problems involving grids with solar power plants or panels, accurately forecasting the power output of a solar power plant is of critical importance. This forecast serves as input for the environment model used by the ML (or RL) agent to learn system control [30]. In essence, accurate future predictions lead to the creation of reliable rewards, enabling the agent to learn more effective control strategies [31].

The comparison table of the models we have prepared based on all the studies that we have examined within the scope of making this study and the model we propose is given in Table 2. It is seen that almost all of the solar power forecasting studies mentioned or not mentioned in this section require a meteorological input. Additionally, in some studies, it is seen that labels are created with certain intervals from the generated power and DL models using the input-output relationship are developed. With these in mind, we can list the main contributions of this paper to the literature as follows. This study:

- Proposes a novel solar power forecasting model based on the NeuralProphet algorithm, which eliminates the dependency on meteorological and other external datasets, simplifying the forecasting process.
- Demonstrates the effectiveness of the proposed method through two case studies at different scales: a small-scale dataset collected at Bursa Technical University and a

large-scale dataset representing nationwide solar power generation in Germany.

- Achieves high predictive accuracy, with R-squared values exceeding 0.99 for large-scale forecasts, showcasing the capability of NeuralProphet to capture strong seasonal patterns and trends in solar power generation data.
- Introduces a data-independent forecasting methodology that can be adapted to diverse scenarios, paving the way for more scalable and accessible renewable energy forecasting solutions.

In contrast to these data-intensive and often computationally complex approaches, our proposed NeuralProphet-based method is uniquely characterized by its independence from meteorological or exogenous input data. While models like LSTM and GRU achieve high accuracy, they typically rely on diverse meteorological inputs, making them less suitable for data-scarce or infrastructure-limited scenarios. NeuralProphet, on the other hand, captures seasonality, trends, and autoregressive features directly from historical generation data. This simplifies deployment, reduces input complexity, and significantly improves applicability, especially in real-world settings where data availability is limited.

III. DATA ACQUISITION AND DATASET CREATION

The data comprises power generation data from a grid-connected PV system. These data were recorded at the Renewable Energy Systems and Smart Grids Laboratory of Bursa Technical University (40.1828° N, 29.0665° E). Hourly data, with mean values aggregated every 15 minutes, were utilized. The data cover the period from March 1, 2024, to November 30, 2024.

A. EXPERIMENTAL SETUP

Figures 2.a and 2.b showcase the laboratory's exterior and interior setups, respectively, highlighting the comprehensive design of the grid connected-PV system [33], [34]. The PV array comprises six Jiangyin HR-200W-24V solar panels connected in series, each with a 200 Wp capacity, yielding a total system output of 1.2 kWp. Inside the laboratory, key components include the PV electric connection panel, which facilitates secure power distribution, and an SMA SunnyBoy SB 1.3 kWp inverter, which optimizes DC-to-AC power conversion using Maximum Power Point Tracking (MPPT) and features an H5 topology for enhanced efficiency and performance. Additionally, a monitoring-protection box ensures real-time system oversight and safety, while the SMA Data Manager M and SMA Sensorbox enable data management and precise environmental monitoring. During data collection from the experimental system, the inverter's output electrical data is transmitted to the Data Manager via an RS232 connection. The Data Manager, connected to the network via Ethernet, transmits the data in real-time to the PC, where the power generation data from the PV system is monitored and recorded.

TABLE 2. Comparison of all major ML and DL based solar power forecasting methods in the literature with the proposed approach.

Model	Temporal Resolution	Data Preprocessing	Compt. Efficiency	Dependency*	Accuracy	Scalability
NeuralProphet	High	Moderate	High	Low	Excellent	Excellent
Prophet [26]	High	Low	High	Low	Good	Moderate
LSTM [4], [7], [24], [20], [22]	Moderate	High	Moderate	High	Very Good	High
RNN [24]	Moderate	High	Moderate	High	Good	High
CNN-LSTM [11], [18], [19]	High	High	Low	High	Excellent	High
GRU [6], [24]	Moderate	High	Moderate	High	Very Good	High
BiLSTM [14], [24]	High	High	Low	High	High	High
LightGBM [15], [16], [32], [17]	High	Moderate	Very High	Moderate	Very Good	Very High
SVM [1], [1], [3], [13], [21]	Moderate	High	Low	High	Good	Low

*Dependency on External Feature

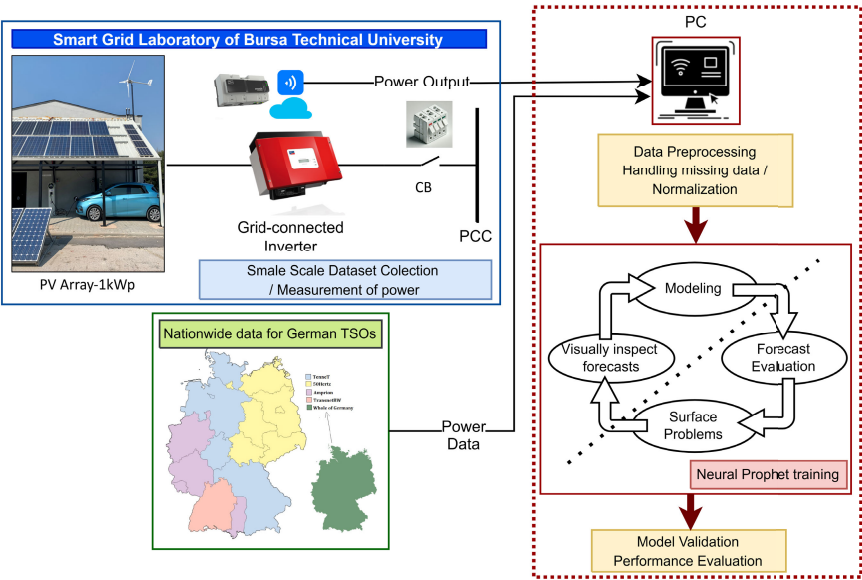


FIGURE 1. The methodology for solar power prediction using proposed algorithm.

B. MEASUREMENT OF POWER OUTPUT

The measured data has a resolution of 5 minutes when recorded daily and 15 minutes when recorded weekly. The power generation data from the system in the laboratory for the week of November 12 to November 18, 2024, is presented in Fig. 3. The graph shows the power output of the PV system over a series of days in October 2024. Peaks in power generation are observed on sunny days, with values reaching up to 1 kWp, while cloudy days show lower and more fluctuating output. This trend highlights the variability in PV energy production due to weather conditions.

Fig. 4. illustrate the daily power output of the PV system on two different days, showcasing the impact of weather conditions on solar generation. This comparison highlights the system’s sensitivity to atmospheric conditions, with stable solar radiation resulting in steady power output, while intermittent cloud cover leads to significant variability.

The dataset used in this study is one of its key contributions and distinguishes it from other works in the literature. We collected real-time solar power generation data over a period of ten months using a system of large-size solar panels installed at the Smart Grid Laboratory of Bursa Technical University. Data was recorded at two different intervals: every

five minutes for daily datasets and every fifteen minutes for weekly datasets. This real-world data collection effort enabled the development of a dataset that closely reflects actual power production conditions, rather than simulated or third-party provided datasets. Fig. 5. shows the solar power generation distribution of the smart grid lab at BTU.

Before the data could be used to train the NeuralProphet models, it underwent a thorough data preprocessing stage to ensure its suitability for solar forecasting. This preprocessing involved several key steps. Firstly, outliers in the data were identified and removed. These outliers could be the result of data recording errors, extreme and unrepresentative consumption spikes, or other anomalies that could skew the model’s learning process and negatively impact the accuracy of the forecasts. Secondly, given the nature of real-world datasets, some missing values were expected. These gaps were addressed by appropriate data imputation methods, ensuring the continuity and consistency of the time series data without introducing bias. Moreover, the time series data was carefully aligned to ensure that all input features corresponded to the correct time intervals, which is crucial for accurate forecasting. The use of real, high-resolution data collected at 15-minute intervals not only enhances



FIGURE 2. Laboratory's (a) (above) exterior and (b) (below) interior setups.

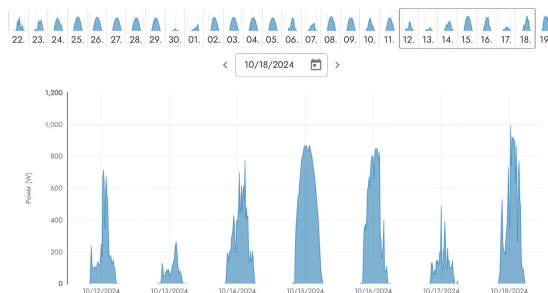


FIGURE 3. Daily variations in power output reflecting weather conditions.

the accuracy of the forecasting models but also makes the findings of this study highly applicable to real-world scenarios.

C. COUNTRYWIDE DATASET: CASE STUDY DATASET ANALYSIS

The paper presents two forecasting studies, small-scale and large-scale, to demonstrate the success of the proposed forecasting approach. The dataset that we collected ourselves and described above how we created the dataset is chosen as small-scale. The large-scale forecasting exercise focuses

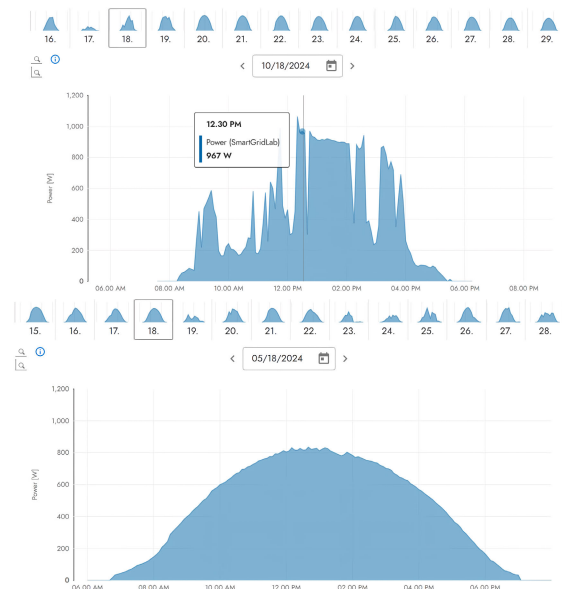


FIGURE 4. Daily distribution characteristic of solar power generation in a rainy day (above) and a sunny day (below).

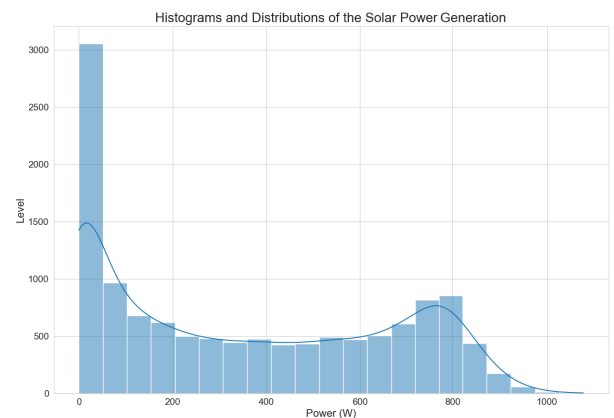


FIGURE 5. Histogram and distribution of solar power generation.

on solar power generation across the country. On the other hand, in order for the models using the proposed NeuralProphet algorithm to predict the desired output, it is essential to analyze the trends and seasonality of the output over time. In this context, our own data, for which we provide results in the small scale category, are promising but may be insufficient. In this context, a large-scale nationwide analysis was conducted to demonstrate the power of the proposed method. Germany was chosen as a case study. Certain power distribution companies operate in Germany. TenneT, Amprion, TransnetBW and 50Hertz are the operators responsible for Germany's TSOs. These companies are responsible for ensuring the operation of high-voltage power grids and maintaining the reliability of energy supply. The electricity transmission system in Germany and Europe is organized regionally, with each company operating in a specific geographical area. TenneT is the operator responsible for electricity transmission in the northern regions of Germany and the Netherlands. The

company is particularly involved in pioneering work on the integration of offshore wind energy projects into the grid. Amprion plays an important role in the energy transition by providing power transmission in the western regions of Germany, especially in industrial areas. TransnetBW is active in the southwest of Germany, especially in the state of Baden-Württemberg, and focuses on the integration of renewable energy sources. 50Hertz is responsible for electricity transmission in the eastern regions of Germany and Hamburg and specializes in the integration of wind energy into the grid. It also contributes to international cooperation by providing energy connections with neighboring countries such as Denmark and Poland. The areas of activity of these TSOs in Germany can be seen in Figure 6.

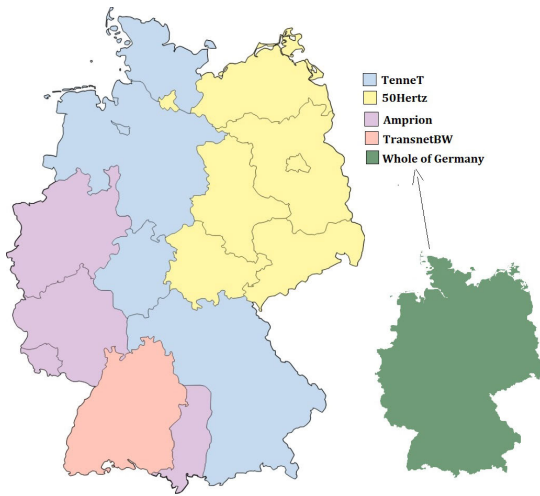


FIGURE 6. Areas of activities for TSOs in germany.

These companies carry out large-scale energy projects and work for the integration of Europe's energy networks to help Germany achieve its energy transition goals. They are also members of the international organization ENTSO-E (European Network of Transmission System Operators for Electricity) [35], which was established to harmonize and coordinate transmission system operators across Europe. The open power system data platform has organized the power generation data provided by ENTSO-E into datasets at different intervals and makes them available to researchers [36]. We selected solar power generation data collected over 5 years and at 15-minute intervals to train our proposed approximation-based forecasting models. Before the models, we preprocessed the data and correctly filled in the missing data, as well as extracted outlier data. Using the datasets generated by ENTSO-E, the models using the proposed method are trained separately and the results are obtained for the prediction of solar power generation data in the operating area of these four major companies. In addition, a separate model is trained and results are obtained for estimating the solar power to be generated in Germany with the solar power generation data of the whole of Germany.

IV. DEMYSTIFYING NEURALPROPHET AND PROPOSED ALGORITHM

In this section, we delve into the mathematical foundations and methodological approach underlying the NeuralProphet model, a sophisticated DL framework designed specifically for time series forecasting.

A. MATHEMATICAL FOUNDATIONS OF NEURALPROPHET

NeuralProphet is a hybrid forecasting framework that builds upon the well-established Prophet model, incorporating deep learning capabilities for enhanced accuracy and flexibility. This model retains the interpretability of classical time series methods while extending their applicability through modular design and neural network-based components. This paper presents the mathematical foundation, innovations, and performance results of NeuralProphet, demonstrating its significant improvements in forecasting accuracy and explainability across diverse datasets [37].

Forecasting time series data is critical for decision-making across industries such as energy, retail, and finance. NeuralProphet introduces a modular architecture where each component contributes additively or multiplicatively to the forecast. The total forecast at time t is defined as:

$$\hat{y}_t = T(t) + S(t) + E(t) + F(t) + A(t) + L(t), \quad (1)$$

where $T(t)$ is the trend, $S(t)$ represents seasonal effects, $E(t)$ accounts for event and holiday impacts, $F(t)$ models known future regressors, $A(t)$ captures auto-regressive effects, and $L(t)$ represents lagged regressors.

The trend component is modeled as a piecewise linear function with changepoints:

$$T(t) = (\delta_0 + \Gamma(t)^\top \delta) \cdot t + (\rho_0 + \Gamma(t)^\top \rho), \quad (2)$$

where $\Gamma(t)$ is a binary vector indicating changepoints, δ_0 is the initial growth rate, and ρ_0 is the initial offset.

Seasonality is captured using Fourier terms:

$$S_p(t) = \sum_{j=1}^k \left(a_j \cos\left(\frac{2\pi jt}{p}\right) + b_j \sin\left(\frac{2\pi jt}{p}\right) \right), \quad (3)$$

with p being the periodicity and k the number of terms.

The auto-regressive effects, crucial for short-term forecasting, are modeled as:

$$\begin{aligned} A(t), A(t+1), \dots, A(t+h-1) \\ = \text{AR-Net}(y_{t-1}, y_{t-2}, \dots, y_{t-p}), \end{aligned} \quad (4)$$

where p is the order of the AR model.

Future and lagged regressors are integrated as follows:

$$L(t) = \sum_{x \in X} L_x(x_{t-1}, x_{t-2}, \dots, x_{t-p}), \quad (5)$$

$$F(t) = \sum_{f \in F} F_f(t). \quad (6)$$

The model accommodates events and holidays through binary variables, modeled as:

$$E(t) = \sum_{e \in E} E_e(t). \quad (7)$$

1) TRAINING AND OPTIMIZATION

NeuralProphet is implemented in PyTorch, enabling flexible and scalable training with stochastic gradient descent (SGD). The default loss function is the Huber loss, which balances sensitivity to outliers:

$$L_{\text{huber}}(y, \hat{y}) = \begin{cases} \frac{1}{2\beta}(y - \hat{y})^2, & \text{if } |y - \hat{y}| < \beta \\ |y - \hat{y}| - \frac{\beta}{2}, & \text{otherwise.} \end{cases} \quad (8)$$

Regularization is applied to prevent overfitting, using a scaled log transform of weights:

$$\Lambda(\theta, \epsilon, \alpha) = \frac{1}{n} \sum_{i=1}^n \log \left(\frac{1}{\epsilon e + \alpha |\theta_i|} \right) + \log(\epsilon) + 1. \quad (9)$$

NeuralProphet demonstrates substantial improvements over Prophet in both synthetic and real-world datasets. On datasets with complex dynamics, the model achieves up to 92% better forecasting accuracy for short to medium-term horizons. Its modular design ensures interpretability, allowing domain experts to review and adjust forecasts when necessary. Compared to Prophet, NeuralProphet is slower in training but significantly faster during prediction, making it ideal for real-time applications. Furthermore, its ability to integrate auto-regressive components and lagged regressors extends its applicability to datasets with high temporal dependencies.

B. PROPOSED ALGORITHM FOR SOLAR FORECASTING

NeuralProphet bridges the gap between classical and machine learning-based forecasting by combining the interpretability of traditional methods with the scalability and flexibility of neural networks. Its modular framework, state-of-the-art training procedures, and robust performance across diverse applications make it a powerful tool for time series forecasting. The advantages of the NeuralProphet method, as proposed in this study, lie in its ability to decompose time series data into its fundamental components—trend, seasonality, and residuals—allowing for a more nuanced understanding of energy demand patterns. Unlike traditional statistical methods and other ML and DL models, NeuralProphet excels in handling multiple seasonality and incorporates autoregressive terms, enhancing its capacity to model both long-term and short-term dependencies in the data. This flexibility and adaptability make it particularly well-suited for solar forecasting in various scales.

Furthermore, NeuralProphet is particularly well-suited for forecasting tasks where the data exhibits strong seasonal patterns and where external factors or events significantly influence the outcome. In Solar forecasting, NeuralProphet's

ability to model complex seasonal patterns and incorporate external variables makes it an ideal choice for accurate, reliable energy productions predictions. The pseudocode of the algorithm proposed in this study for solar forecasting at different scales is given in “algorithm 1”.

Algorithm 1 Proposed NeuralProphet-Based Solar Forecasting

```

0: Input: Dataset  $D$ , Prediction horizon  $H$ , Parameters
0: Output: Trained model  $M_C$ , Performance metrics
0: procedure Preprocessing( $D$ )
0:   Remove outliers and fill missing values in  $D$ 
0:   Normalize data:  $\tilde{D} \leftarrow \frac{D - \mu_D}{\sigma_D}$ 
0: end procedure
0: procedure HyperparameterTuning( $D_C$ )
0:   Set hyperparameters  $P_C$  based on seasonality  $\mathcal{S}$ ,
   growth  $\mathcal{G}$ , and autoregression  $\mathcal{A}$ 
0:    $\mathcal{S}_C(t) \leftarrow \sum_{k=1}^K \alpha_k \cos\left(\frac{2\pi kt}{T}\right) + \beta_k \sin\left(\frac{2\pi kt}{T}\right)$ 
0:    $\mathcal{G}_C(t) \leftarrow g(t; \mathbf{a}_C)$ 
0:    $\mathcal{A}_C(t) \leftarrow \sum_{p=1}^P \phi_p y_{t-p}$ 
0: end procedure
0: procedure TrainModel( $\tilde{D}_C, P_C$ )
0:   Initialize model  $M_C$  with parameters  $\theta_C = \{\mathcal{S}_C, \mathcal{G}_C, \mathcal{A}_C\}$ 
0:   while not converged do
0:     Update  $\theta_C$  by minimizing loss  $L(\theta_C; \tilde{D}_C) = \sum_{t=1}^T (y_t - \hat{y}_t)^2$ 
0:   end while
0: end procedure
0: procedure Predict( $M_C, H$ )
0:   Generate predictions  $\hat{y}_C(t) = \mathcal{S}_C(t) + \mathcal{G}_C(t) + \mathcal{A}_C(t)$ 
   for  $t = T + 1$  to  $T + H$ 
0: end procedure
0: procedure Evaluate( $\hat{y}_C, y_C$ )
0:   Calculate  $R^2 = 1 - \frac{\sum_{t=1}^T (y_t - \hat{y}_t)^2}{\sum_{t=1}^T (y_t - \bar{y})^2}$ 
0:   Calculate error metrics: MSE, RMSE, MAE, SMAPE
0: end procedure
0: procedure SaveModel( $M_C$ )
0:   if  $R^2$  and accuracy metrics satisfactory then
0:     Save model  $M_C$ 
0:   end if
0: end procedure

```

In this study, the application of NeuralProphet to solar forecasting leads various advantages. Main advantages of the proposed methods is freeing from methological dependecises.

C. MODEL EVALUATION

To evaluate the performance of the proposed models, we utilized the following metrics: R-squared (R^2), MAE, RMSE, Mean Bias Error (MBE), and Symmetric Mean Absolute Percentage Error (SMAPE). These metrics provide a comprehensive understanding of the models' predictive

TABLE 3. Utilized DL model evaluation methods.

Evaluation Method	Calculation Formula
Mean absolute error (MAE)	$MAE = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $
Root mean squared error (RMSE)	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
Mean bias error (MBE)	$MBE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$
Symmetric Mean Absolute Percentage Error	$SMAPE = \frac{100\%}{N} \sum_{i=0}^{N-1} \frac{2 * y_i - \hat{y}_i }{ y_i + \hat{y}_i }$
R-Squaed (R^2)	$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$

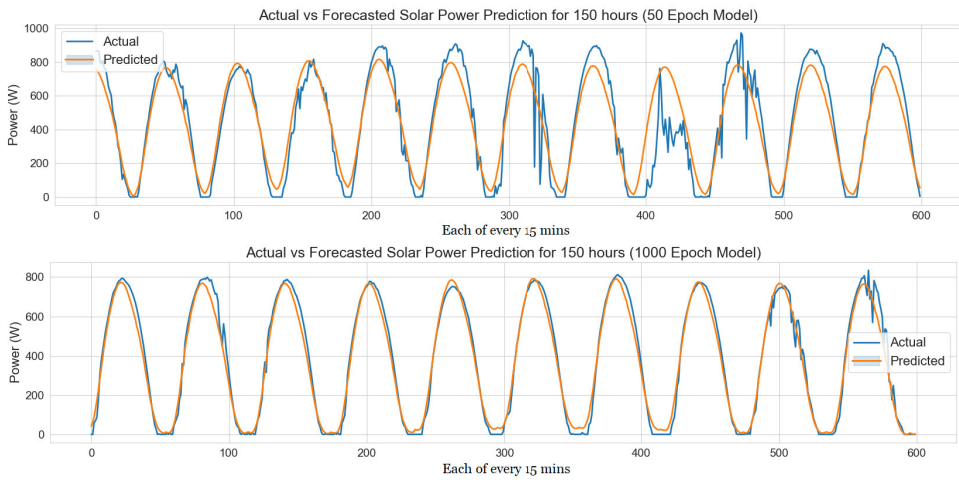


FIGURE 7. Actual versus predicted solar power generation using the NeuralProphet model in the small-scale experimental setup (BTU dataset). The x-axis denotes time (in 15-minute intervals), and the y-axis denotes solar power (in watts). The orange line represents NeuralProphet’s predicted values, while the blue line represents the actual recorded values for 50-epoch and 1000-epoch models as cases.

capabilities and their alignment with actual values. The formulas for these metrics are presented in Table 3 for reference.

R^2 metric indicates the proportion of the variance in the observed data that is explained by the predictive model. An R^2 value closer to 1 signifies a strong correlation between the predicted and actual values, reflecting the model’s ability to capture underlying trends and seasonality in the data. MAE measures the average magnitude of errors between predicted and actual values, regardless of their direction. It provides a straightforward interpretation of the typical error magnitude, helping to evaluate the model’s accuracy in practical terms. Besides, RMSE is a metric that penalizes larger errors more heavily compared to MAE. By squaring the errors before averaging and taking the square root, RMSE highlights significant deviations between predictions and actual values, making it particularly useful for understanding the impact of extreme outliers. Moreover, MBE quantifies the average bias

in the predictions by measuring the mean difference between predicted and actual values. A positive MBE indicates an overestimation by the model, while a negative value reflects underestimation. This metric sheds light on systematic biases in the model. On the other hand, SMAPE is a normalized metric that evaluates the relative error between predicted and actual values. By dividing the absolute error by the average of the predicted and actual values, SMAPE provides a scale-independent measure of accuracy. This is particularly useful for comparing predictions across datasets with varying magnitudes.

Table 3 summarizes the formulas for these metrics, which collectively offer a multidimensional perspective on model performance. By examining these metrics, we can assess not only the overall accuracy of the models but also their ability to handle data variability, identify biases, and manage outliers effectively. This comprehensive evaluation ensures that the proposed forecasting approach meets the

practical requirements of solar power prediction at different scales.

Where: y_i represents actual data and $\hat{y}_i$ represents predicted values for index i . n is the total number of observations in the (public/private) data set, p_i is your target prediction and a_i the actual target for i . The R^2 is calculated by dividing the sum of squares of residuals from the regression model (given by SS_{RES}) by the total sum of squares of errors from the average model (given by SS_{TOT}) and then subtracting it from 1. The result is between zero and one, and the closer it is to one, the better the model yielded. In addition, as a result of multiplying the value of R^2 by 100, the percentage value of precision is obtained.

V. PERFORMANCE RESULTS AND ANALYSIS

Models have been developed and trained using the algorithm proposed in this paper, which provides meteorological and other input-independent predictions for solar power generation. As mentioned earlier, small-scale solar power generation forecasting and large-scale solar power generation forecasting models are developed to emphasize the power of the proposed method. The small-scale model uses real solar power generation data collected by the BTU smart grid laboratory, while the large-scale model is based on a nationwide solar power generation forecasting study. In this section, the first subsection presents the results and performance analysis of the model we trained with the data we collected in our own laboratory, while the second subsection presents the results of the Germany-wide solar power generation forecasting study selected as a case study.

A. SMALL-SCALE SOLAR POWER PREDICTION

We performed forecasting studies with our proposed forecasting algorithm using our own solar power generation data that we are currently collecting. At the time of the study, we had a dataset of about eight months of generation data collected at 15 minute intervals. We were aware that this data set, which had not completed one year, was not sufficient for the actual performance of our proposed seasonality-based forecasting model. For this reason, we created forecasting models with different parameters and analyzed the changes in the forecast outputs. We created five different forecasting models and obtained results. Figure 7 shows the actual and forecasts of the models with the worst and best parameters for 150 hours. to better observe the performance of the models, we excluded the night hours when there was no generation.

The model with a lower number of epochs but a faster training time is sometimes more or less accurate than the actual values. This problem is somewhat overcome by the model with a high number of epochs but a long training time. However, both models are not highly successful in tracking the sharp production changes on cloudy days. The training time is directly proportional to the increase in the number of epochs. The learning performance and error metrics of the small-scale forecasting model trained with different number of epochs are given in Table 4.

TABLE 4. Performance metrics of small-scale solar power forecasting models with varying epoch numbers.

Models	R^2	MAE	RMSE	MBE	SMAPE
50Epoch	0.6988	103.33	162.23	13.335	0.3502
100Epoch	0.7297	102.52	156.55	11.177	0.3372
200Epoch	0.7403	102.10	155.32	10.548	0.3362
500Epoch	0.7423	102.32	155.47	9.4711	0.3369
1000Epoch	0.7451	102.16	155.26	9.2527	0.3367

Table 4 presents the performance results of the NeuralProphet-based forecasting models trained on small-scale solar power generation data collected from the Smart Grids laboratory at BTU. The table shows how model performance varies with different epoch numbers, highlighting key evaluation metrics such as R^2 , end errors.

The R^2 values indicate how well the model explains the variance in the data, with higher values suggesting better predictive capability. As the number of epochs increases, R^2 values gradually improve, reaching 0.7451 for the model trained with 1000 epochs. This suggests that the model becomes better at capturing underlying trends and seasonality in the dataset with extended training. However, gains in R^2 beyond 200 epochs are marginal, implying diminishing returns on performance improvements as training iterations increase.

MAE and RMSE provide insight into the average magnitude of errors. While MAE measures the mean absolute difference between predicted and actual values, RMSE penalizes larger errors more heavily, making it particularly useful for assessing the impact of extreme deviations. The RMSE values show a steady decline with increasing epochs, reinforcing the idea that longer training improves prediction accuracy. However, the slight decrease in MAE and RMSE between 500 and 1000 epochs suggests that excessive training does not significantly enhance prediction accuracy, making 200 epochs a reasonable trade-off between accuracy and computational efficiency.

MBE quantifies the average bias in predictions. Positive values indicate overestimation, while negative values suggest underestimation. The results in Table 3 show that as training epochs increase, the MBE values decrease, suggesting that higher epoch counts reduce systematic biases in the predictions. Finally, SMAPE provides a relative measure of error by normalizing absolute errors, making it useful for comparing models across datasets of varying scales. The nearly stable SMAPE values across all models confirm that the proposed forecasting approach maintains a consistent level of relative accuracy. The training times of the models are given in Table 5.

TABLE 5. Execution time comparison of the models for small-scale case.

Time	50Ep.	100Ep.	200Ep.	500Ep.	1000Ep.
Minutes:	≈4.4	≈12	≈32	≈107	≈196

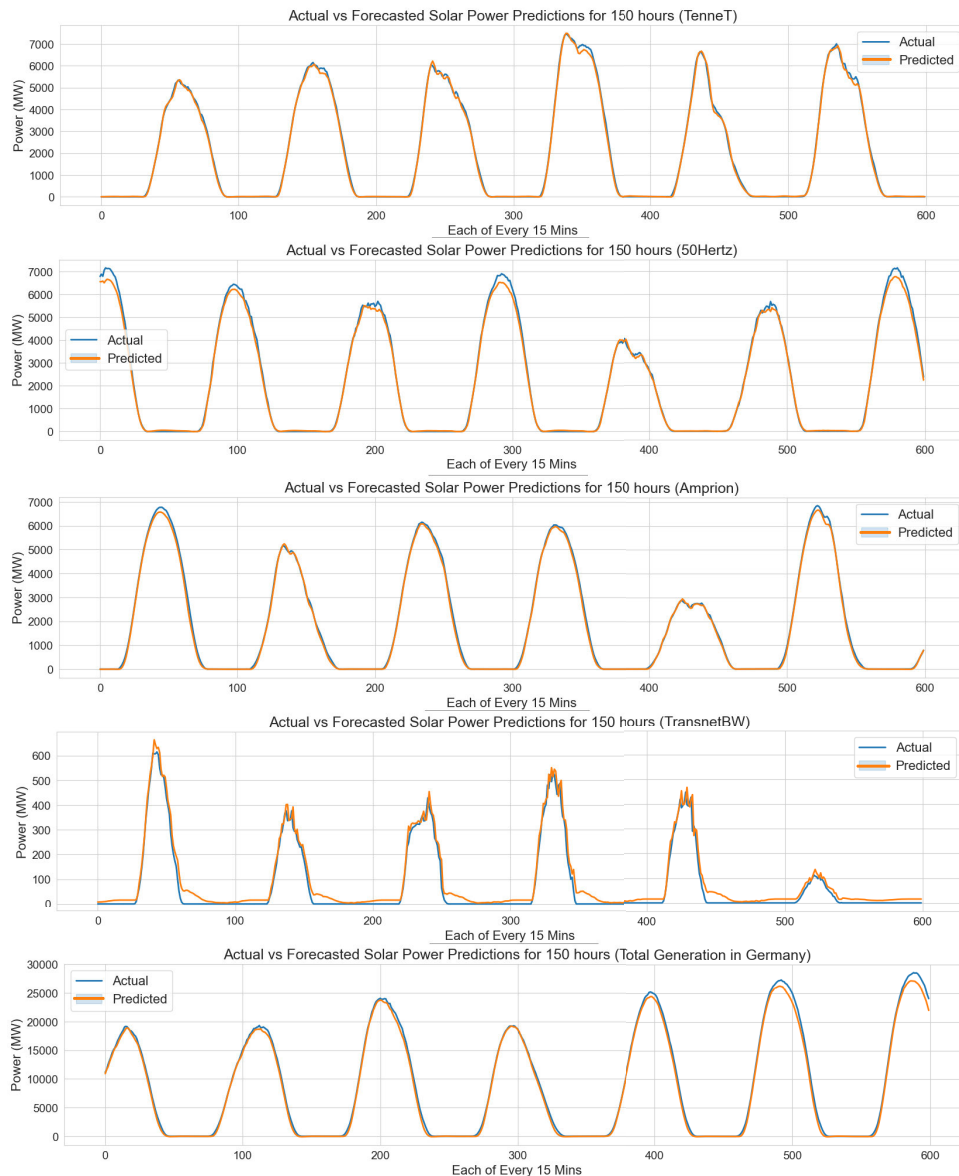


FIGURE 8. Actual versus predicted solar power generation using the NeuralProphet model in the large-scale forecasting case (Germany – TSO-level data). The x-axis denotes time (in 15-minute intervals), and the y-axis denotes solar power (in MW). The graph illustrates the model's strong correlation with real-world generation values, especially over longer horizons.

When the two tables are analyzed, the models with 500 and 1000 epochs are not efficient in terms of training times and success percentages (R^2 score). On the other hand, the model with 200 epochs has a prediction success rate of over 74% and a reasonable training time. With the 1000 epoch model, the most successful forecasting model, historical forecasts are realized. Forecasts were obtained for all training dataset lengths and compared with the actual values. The kernel density estimation outputs of the real and forecast data are given in Fig. 9.

Fig. 9 shows that the densities of the predictions produced by the model using the proposed prediction algorithm are very similar to the densities of the actual values. The prediction model has produced peaks similar to the peaks where the

real values are concentrated. This is an indication that the proposed model shows considerable success even when training with a limited length dataset.

B. LARGE-SCALE SOLAR POWER PREDICTION

The proposed NeuralProphet based algorithm learns the trend and seasonality in the dataset. For this reason, recurring hours, days or even months are needed. It is obvious that the proposed model shows considerable success after training with our own collected data, which is determined as a small-scale case. However, in order to show how powerful and successful the model can be, a large-scale case study was conducted with five years of data (details of the dataset are given in section III-C). Five forecasting models were

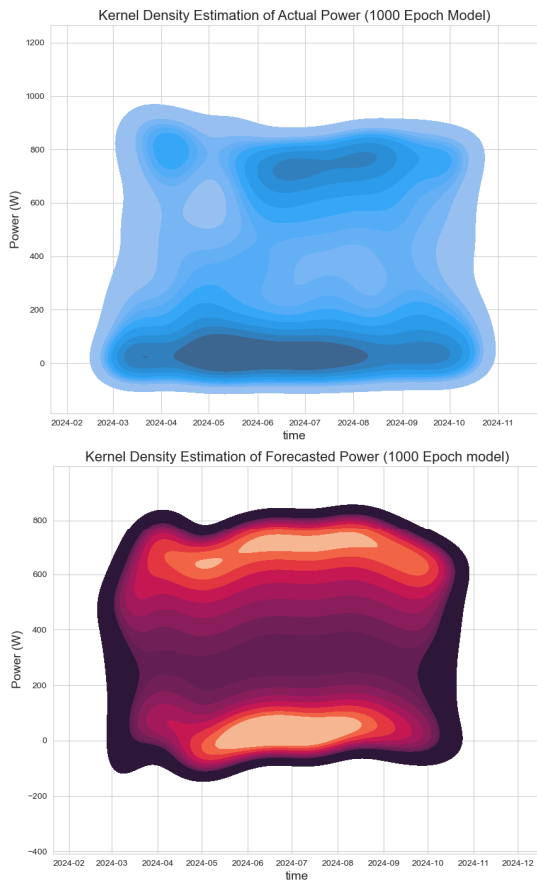


FIGURE 9. Kernel density estimation graphs of actual and predicted solar power values for BTU smart grid lab. Dataset as small-scale.

developed and trained with data from four German TSOs and nationwide data. Figure 10 shows how the forecasting model developed for Germany as a whole reduced the loss values used to optimize itself during training and validation. It reduced the loss value to a very small value and converged, which indicates successful training performance.

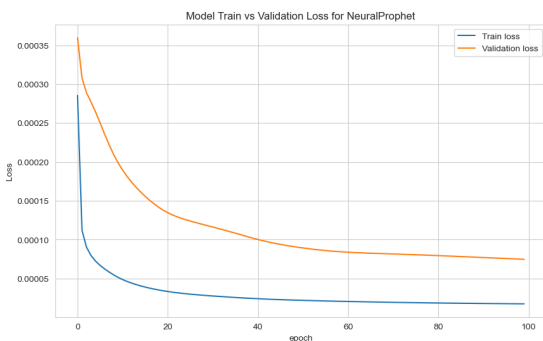


FIGURE 10. Learning performance of total germany prediction model (training:blue, validation:orange).

Prediction values that closely follow the actual production values are an indicator of success. In this context, Fig. 8 shows how close the trained models for the four TSOs and for Germany as a whole are to the actual values. R^2 values

and error metric outputs, which are other success indicators of the models, are given in Table 6.

TABLE 6. Performance metrics of large-scale solar power forecasting models trained on TSOs' datasets.

Training	R^2	MAE	RMSE	MBE	SMAPE
<i>TenneT</i>	0.9965	101.90	160.72	-16.208	0.529
<i>Amprion</i>	0.9985	49.989	73.897	11.655	0.528
<i>50Hertz</i>	0.9936	93.232	166.72	-33.401	0.531
<i>TransnetBW</i>	0.9981	26.257	43.216	1.873	1.000
<i>Whole Germany</i>	0.9979	202.19	329.16	-38.976	0.519

Table 6 presents the performance results of the proposed forecasting models trained with large-scale solar power generation data from German TSOs. The findings demonstrate significantly higher R^2 values compared to the small-scale dataset, with all models achieving R^2 values above 0.99. This indicates that the large-scale models capture seasonal patterns and trends with remarkable accuracy, benefiting from the extensive five-year dataset.

The differences in performance among TSOs can be attributed to variations in their respective operating regions, which influence the consistency of solar power generation. The model trained with Amprion's dataset achieves the highest R^2 (0.9985) and the lowest RMSE (73.897), suggesting that solar power generation in this region follows highly predictable patterns. In contrast, the model trained with TransnetBW's data achieves the lowest RMSE but the highest SMAPE, indicating that while absolute errors are minimized, relative errors remain more pronounced due to the variability in the dataset.

The Whole Germany model, trained on aggregated solar power generation data, achieves an R^2 of 0.9979, demonstrating its capability to generalize well across the entire country. However, it exhibits a relatively higher MAE and RMSE compared to individual TSOs, likely due to greater variability in national-level solar power generation patterns. The negative MBE for TenneT, 50Hertz, and Whole Germany suggests a slight tendency toward underestimation in those regions, whereas the positive MBE for Amprion and TransnetBW indicates minor overestimation. Using this successful model, a six-month historical forecast was made for the whole of Germany. Then, the densities of the actual and forecast values are presented (see Fig. 11).

The most successful model among all models was the Amprion forecasting model. This model not only achieved the highest R^2 score but also managed to minimize the error metrics between the actual and forecast values. The accuracy of this model can be seen from the similarity of the statistical distribution of the actual and forecast data sets. Figure 12 shows these similar statistical distributions.

Overall, the results suggest that the NeuralProphet-based forecasting approach is highly effective, particularly at a large scale where strong seasonal trends enhance model predictability. While increased training epochs contribute to improved accuracy, the returns diminish beyond a certain threshold, making it crucial to balance computational

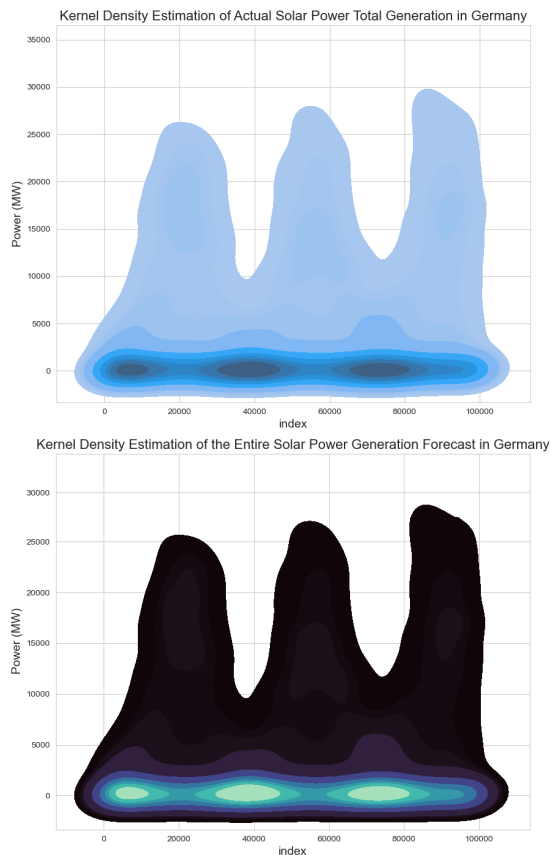


FIGURE 11. Kernel density estimation graphs of actual and predicted total solar power values of Germany as large-scale case.

efficiency with performance. The findings highlight the robustness of the proposed method in different forecasting scenarios and emphasize its applicability for scalable solar power prediction across varying operational contexts. The nationwide solar power generation forecasting study conducted as a large-scale forecasting study demonstrates the success of the proposed model. Both the 15-minute forecasts of the solar power to be generated by the whole of Germany and the solar power in the areas of the TSOs operating in this country have been successfully realized thanks to the proposed model. This model, which is promising in the small-scale case, which we have collected for ten months, proves its success in large-scale forecasting. To further assess the effectiveness of our model, we compared its forecasting performance with typical ranges reported in literature for advanced ML/DL methods. For example, LSTM and CNN-LSTM architectures generally achieve R^2 values ranging between 0.85 and 0.95 for day-ahead forecasts when trained with high-quality meteorological data [4], [7], [10], [20]. Our model, without relying on such exogenous inputs, achieves R^2 values exceeding 0.99 in the large-scale Germany case and up to 0.745 in the small-scale case. These results suggest that NeuralProphet, when appropriately tuned, can match or even exceed the performance of complex multi-input models in

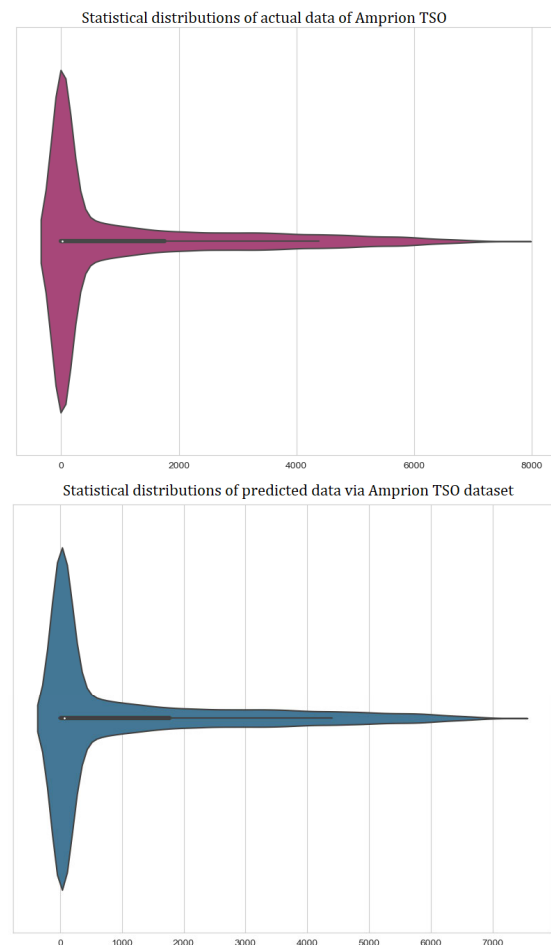


FIGURE 12. Statistical characteristic graphs of actual and predicted solar power values for Amprion TSO dataset as large-scale case.

certain scenarios, with the added benefit of lower complexity and enhanced portability.

VI. CONCLUSION

This study introduces a novel approach to forecasting solar power generation by exploiting the NeuralProphet algorithm. Unlike traditional methods that rely on external datasets such as meteorological or sensor data, the proposed method predicts solar power generation solely based on historical generation data. By eliminating the dependency on additional inputs, this approach simplifies the forecasting process while maintaining competitive accuracy, particularly in data-limited scenarios.

The two primary contributions of this paper to the literature are the development and demonstration of a new forecasting algorithm based on the NeuralProphet method, and the application of the model to both small-scale and large-scale datasets, encompassing both our own collected data and publicly available real-world data. These contributions address key challenges in solar power forecasting by introducing a scalable, data-independent methodology and validating its performance across diverse scenarios.

In the first case study, the proposed algorithm was applied to a 10-month dataset from a small-scale solar power generation unit. Despite the limited data size, the model achieved an R-squared value above 0.74, demonstrating its ability to identify patterns and trends in solar power generation. The second case study focused on large-scale solar power generation in Germany, where the algorithm achieved R-squared values exceeding 0.99 across datasets from four major TSOs and the nationwide dataset. These results highlight the algorithm's robustness and its ability to capture the strong seasonality inherent in solar power generation, even at a national scale. By presenting this new approach, the study makes a significant contribution to the literature on solar power forecasting. The demonstrated success of the NeuralProphet-based algorithm suggests its potential applicability to a wide range of forecasting problems, including other renewable energy sources or scenarios with limited auxiliary data.

To better contextualize the performance of our proposed model, we compared our results against commonly reported benchmarks in the solar forecasting literature. Studies using LSTM and CNN-LSTM architectures typically report R^2 values ranging from 0.85 to 0.95 for short-term solar generation forecasts when trained with high-quality meteorological and irradiance data. MAE values in these studies often range between 50 MW and 202 MW (around 102 W in small-scale case), while RMSE values commonly fall within the 43 MW to 330 MW (around 155 W in small-scale case) range depending on the scale and complexity of the forecasting task. In this context, our model's performance is highly competitive: despite relying solely on historical power data without any exogenous input, it achieved R^2 values of 0.7451 in the small-scale case and above 0.99 in all large-scale (TSO-level) forecasts, with corresponding MAE and RMSE values well within or better than these literature ranges. This strongly supports the effectiveness of NeuralProphet as a practical and data-independent forecasting tool.

Future research can extend the current work in several directions. One promising avenue is the integration of NeuralProphet into hybrid models that combine it with probabilistic forecasting techniques, such as quantile regression or Bayesian deep learning, to quantify uncertainty in solar power predictions. Additionally, multi-step forecasting frameworks could be explored to improve long-horizon prediction performance and reduce cumulative errors in dynamic systems. Another interesting extension involves adapting the model to other forms of renewable generation (e.g., wind, hydro) to evaluate its generalizability. Finally, future studies could test the model in climatically diverse regions, assess its performance under extreme weather scenarios, and investigate the use of attention mechanisms to enhance temporal feature learning in NeuralProphet.

In conclusion, this research advances the field of solar power generation forecasting by providing a practical and innovative solution that achieves accurate predictions while

simplifying the data requirements. This methodology sets the stage for broader applications and invites further exploration of data-independent forecasting techniques in renewable energy and beyond.

REFERENCES

- [1] H. Kim and D. Lee, "Probabilistic solar power forecasting based on bivariate conditional solar irradiation distributions," *IEEE Trans. Sustain. Energy*, vol. 12, no. 4, pp. 2031–2041, Oct. 2021.
- [2] S. Al-Dahidi, O. Ayadi, M. Alrbai, and J. Adeeb, "Ensemble approach of optimized artificial neural networks for solar photovoltaic power prediction," *IEEE Access*, vol. 7, pp. 81741–81758, 2019.
- [3] S. Wen, C. Zhang, H. Lan, Y. Xu, Y. Tang, and Y. Huang, "A hybrid ensemble model for interval prediction of solar power output in ship onboard power systems," *IEEE Trans. Sustain. Energy*, vol. 12, no. 1, pp. 14–24, Jan. 2021.
- [4] M. Mishra, P. Byomakesha Dash, J. Nayak, B. Naik, and S. Kumar Swain, "Deep learning and wavelet transform integrated approach for short-term solar PV power prediction," *Measurement*, vol. 166, Dec. 2020, Art. no. 108250.
- [5] Y. Gao, S. Miyata, and Y. Akashi, "Interpretable deep learning models for hourly solar radiation prediction based on graph neural network and attention," *Appl. Energy*, vol. 321, Sep. 2022, Art. no. 119288.
- [6] H. Yue, M. M. Ali, Y. Lin, and H. Liu, "Ultra-short-term forecasting of large distributed solar PV fleets using sparse smart inverter data," *IEEE Trans. Sustain. Energy*, vol. 15, no. 3, pp. 1968–1980, Jul. 2024.
- [7] M. Chai, F. Xia, S. Hao, D. Peng, C. Cui, and W. Liu, "PV power prediction based on LSTM with adaptive hyperparameter adjustment," *IEEE Access*, vol. 7, pp. 115473–115486, 2019.
- [8] N. Tang, S. Mao, Y. Wang, and R. M. Nelms, "Solar power generation forecasting with a LASSO-based approach," *IEEE Internet Things J.*, vol. 5, no. 2, pp. 1090–1099, Apr. 2018.
- [9] Y. Zhang, M. Beaudin, R. Taheri, H. Zareipour, and D. Wood, "Day-ahead power output forecasting for small-scale solar photovoltaic electricity generators," *IEEE Trans. Smart Grid*, vol. 6, no. 5, pp. 2253–2262, Sep. 2015.
- [10] L. Cheng, H. Zang, Z. Wei, T. Ding, R. Xu, and G. Sun, "Short-term solar power prediction learning directly from satellite images with regions of interest," *IEEE Trans. Sustain. Energy*, vol. 13, no. 1, pp. 629–639, Jan. 2022.
- [11] S. Ghimire, T. Nguyen-Huy, R. C. Deo, D. Casillas-Pérez, and S. Salcedo-Sanz, "Efficient daily solar radiation prediction with deep learning 4-phase convolutional neural network, dual stage stacked regression and support vector machine CNN-REGST hybrid model," *Sustain. Mater. Technol.*, vol. 32, Jul. 2022, Art. no. e00429.
- [12] N. Aksoy and I. Genc, "Design of graphical user interface for artificial intelligence-based energy management system for microgrids," *Electrica*, vol. 23, no. 2, pp. 202–2011, 2023.
- [13] K. Y. Bae, H. S. Jang, and D. K. Sung, "Hourly solar irradiance prediction based on support vector machine and its error analysis," *IEEE Trans. Power Syst.*, vol. 32, no. 2, pp. 935–945, Mar. 2017.
- [14] P. Singla, M. Duhan, and S. Saroha, "An ensemble method to forecast 24-h ahead solar irradiance using wavelet decomposition and BiLSTM deep learning network," *Earth Sci. Inform.*, vol. 15, no. 2, pp. 291–306, 2022.
- [15] N. Aksoy and I. Genc, "Predictive models development using gradient boosting based methods for solar power plants," *J. Comput. Sci.*, vol. 67, Mar. 2023, Art. no. 101958.
- [16] Z. Ma, H. Chang, Z. Sun, F. Liu, W. Li, D. Zhao, and C. Chen, "Very short-term renewable energy power prediction using XGBoost optimized by TPE algorithm," in *Proc. 4th Int. Conf. HVDC (HVDC)*, Nov. 2020, pp. 1236–1241.
- [17] Y. Ju, G. Sun, Q. Chen, M. Zhang, H. Zhu, and M. U. Rehman, "A model combining convolutional neural network and LightGBM algorithm for ultra-short-term wind power forecasting," *IEEE Access*, vol. 7, pp. 28309–28318, 2019.
- [18] J. Ren, Z. Yu, G. Gao, G. Yu, and J. Yu, "A CNN-LSTM-LightGBM based short-term wind power prediction method based on attention mechanism," *Energy Rep.*, vol. 8, pp. 437–443, Aug. 2022.
- [19] S. Lu and T. Bao, "Short-term electricity load forecasting based on neural prophet and CNN-LSTM," *IEEE Access*, vol. 12, pp. 76870–76879, 2024.

- [20] H.-Y. Cheng, C.-C. Yu, and C.-L. Lin, "Day-ahead to week-ahead solar irradiance prediction using convolutional long short-term memory networks," *Renew. Energy*, vol. 179, pp. 2300–2308, Dec. 2021.
- [21] Q. Zhang, H. Yan, and Y. Liu, "Power generation forecasting for solar plants based on dynamic Bayesian networks by fusing multi-source information," *Renew. Sustain. Energy Rev.*, vol. 202, Sep. 2024, Art. no. 114691.
- [22] D. Rangelov, M. Boerger, N. Tcholtchev, P. Lämmel, and M. Hauswirth, "Design and development of a short-term photovoltaic power output forecasting method based on random forest, deep neural network and LSTM using readily available weather features," *IEEE Access*, vol. 11, pp. 41578–41595, 2023.
- [23] S. Tajjour, S. S. Chandel, M. A. Alotaibi, H. Malik, F. P. García Márquez, and A. Afthanorhan, "Short-term solar irradiance forecasting using deep learning techniques: A comprehensive case study," *IEEE Access*, vol. 11, pp. 119851–119861, 2023.
- [24] N. Aksoy and I. Genc, "Deep learning approaches for robust prediction of large-scale renewable energy generation: A comprehensive comparative study from a national context," *Intell. Data Anal., Int. J.*, vol. 29, no. 4, 2025, doi: [10.1177/1088467X251325068](https://doi.org/10.1177/1088467X251325068).
- [25] R. Nelega, D. I. Greu, E. Jecan, V. Rednic, C. Zamfirescu, E. Puschita, and R. V. F. Turcu, "Prediction of power generation of a photovoltaic power plant based on neural networks," *IEEE Access*, vol. 11, pp. 20713–20724, 2023.
- [26] A.-N. Sharkawy, M. M. Ali, H. H. H. Mousa, A. S. Ali, G. T. Abdel-Jaber, H. S. Hussein, M. Farrag, and M. A. Ismeil, "Solar PV power estimation and upscaling forecast using different artificial neural networks types: Assessment, validation, and comparison," *IEEE Access*, vol. 11, pp. 19279–19300, 2023.
- [27] J. Gaboitoalelwe, A. M. Zungeru, A. Yahya, C. K. Lebekwe, D. N. Vinod, and A. O. Salau, "Machine learning based solar photovoltaic power forecasting: A review and comparison," *IEEE Access*, vol. 11, pp. 40820–40845, 2023.
- [28] M. Tasnim, T. I. Rifa, T. Shahriar, and M. A. Habib, "Wind energy deployment in bangladesh: Investigating feasible locations and their characteristics," *Energy Rep.*, vol. 11, pp. 4338–4355, Jun. 2024.
- [29] O. Kirat, A. Çiçek, and T. Yerlikaya, "A new artificial intelligence-based system for optimal electricity arbitrage of a second-life battery station in day-ahead markets," *Appl. Sci.*, vol. 14, no. 21, p. 10032, Nov. 2024.
- [30] N. Aksoy and I. Genc, "Energy storage management for microgrids using n-step bootstrapping," *IEEE Can. J. Electr. Comput. Eng.*, vol. 46, no. 2, pp. 107–116, Spring. 2023.
- [31] E. Foruzan, L.-K. Soh, and S. Asgarpour, "Reinforcement learning approach for optimal distributed energy management in a microgrid," *IEEE Trans. Power Syst.*, vol. 33, no. 5, pp. 5749–5758, Sep. 2018.
- [32] X. Yao, X. Fu, and C. Zong, "Short-term load forecasting method based on feature preference strategy and LightGBM-XGboost," *IEEE Access*, vol. 10, pp. 75257–75268, 2022.
- [33] G. Bayrak, A. Yilmaz, and R. Çakmak, "A new fuzzy&wavelet-based adaptive thresholding method for detecting PQDs in a hydrogen and solar-energy powered EV charging station," *Int. J. Hydrogen Energy*, vol. 48, no. 18, pp. 6855–6870, 2023.
- [34] A. Yilmaz, T. Ateşçi, H. Meral, and G. Bayrak, "A real-time improved ML method for PQD classification of a PV-powered EV charging station," *IEEE Trans. Ind. Electron.*, vol. 72, no. 3, pp. 2622–2632, Mar. 2025.
- [35] *Entso-E, the European Network of Transmission System Operators for Electricity*. Accessed: May 7, 2025. [Online]. Available: <https://www.entsoe.eu/data/transparency-platform/>
- [36] *Open Power System Data*. Accessed: May 7, 2025. [Online]. Available: <https://open-power-system-data.org/>
- [37] O. Triebe, H. Hewamalage, P. Pilyugina, N. Laptev, C. Bergmeir, and R. Rajagopal, "NeuralProphet: Explainable forecasting at scale," 2021, *arXiv:2111.15397*.



data-driven control, robotics, SLAM, path planning, smart grids, and machine learning/deep learning applications in engineering and associated domains.



renowned journals in the last three years. His current research interests include electrical engineering, computer sciences-related areas, such as renewable energy sources and their integration with the grid, real-time automatic fault detection, power quality disturbance classification, islanding detection, and smart grids.



generation, renewable energy, and real-time data acquisition. He has many articles in qualified journals. He has published more than 60 research papers in high-quality peer-reviewed journals and international conferences. He has worked in PV research and renewable energy systems for years and has managed several projects about smart grids.



distributed generation, and SCADA systems. He has experience with horizon and many national electrical grid and distribution projects.

NECATI AKSOY received the B.Sc. degree in electrical and electronics engineering and the Master of Science degree in electrical engineering from Temple University, USA, and the Ph.D. degree from Istanbul Technical University.

He is currently affiliated with the Smart Grid Laboratory. He is also an Assistant Professor with the Department of Electrical and Electronics Engineering, Bursa Technical University. His research interests include reinforcement learning,

ALPER YILMAZ received the M.Sc. degree in electrical-electronic engineering and the Ph.D. degree in electrical engineering from Bursa Technical University, Bursa, Türkiye, in 2018 and 2023, respectively.

He has been working as an Assistant Professor with the Department of Electrical-Electrical Engineering, Bursa Technical University, and a Researcher with the Smart Grids Laboratory, since 2019. He has published nine articles in world-

GÖKAY BAYRAK received the B.S. and Ph.D. degrees from the Electrical-Electronics Engineering Department, Firat University, Türkiye, in 2003 and 2013, respectively.

He is currently an Associate Professor and the Scientific Director of the Smart Grids Laboratory, Department of Electrical-Electrical Engineering, Bursa Technical University. His current research interests include smart grids, AI in energy systems, islanding detection, distributed

MEHMET KOÇ received the B.S. degree from the Electrical-Electronics Engineering Department, Kahramanmaraş Sutcu Imam University, Türkiye, in 1998. He has been an R&D Director of Uludağ Electrical Distribution Company, since 2004. He started the Graduate School, Department of Electrical-Electrical Engineering, Bursa Technical University, in 2024. His current research interests include power systems, distribution systems, renewables, electrification,

...